

Mohammad Ratul Mahjabin

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Education

University of South Florida, Tampa, FL

Ph.D. in Computer Science and Engineering

Jan '24 - Current

Research Areas: Computational Social Science, Human-Computer Interaction, NLP

Advisor: Dr. Raiyan Abdul Baten - [Computational Social Science and AI Lab](#)

Islamic University of Technology, Dhaka, Bangladesh

B.Sc. in Software Engineering

Jan '19 - May '23

Thesis: [An Efficient Deep Learning-based approach for Recognizing Agricultural Pests in the Wild](#)

Publications

Recent preprints under review

- P1 **M. R. Mahjabin** and R. A. Baten, [The Wisdom of Intellectually Humble Networks](#), *arxiv:2502.02015*, 2025

Journal Papers

- J3 Mustafina SN, Islam MN, **Mahjabin MR**, Mannan MM, Islam MM. [Identifying key indicators to develop a novel mobile application for early screening of postpartum depression in developing countries](#). BMC Health Services Research. 2025 Dec;25(1):1-20.
- J2 Rafi TH, **Mahjabin R**, Ghosh E, Ko YW, Lee JG. [Domain generalization for semantic segmentation: a survey](#). Artificial Intelligence Review. 2024 Aug 12;57(9):247.
- J1 Islam MN, Jahangir R, Mohim NS, Wasif-UI-Islam M, Ashraf A, Khan NI, **Mahjabin MR**, Miah AS, Shin J. [A multilingual handwriting learning system for visually impaired people](#). IEEE Access. 2024 Jan 15.

Conference Papers

- C2 Hema NE, Zeest NS, **Mahjabin MR**, Islam MN. [An Ensemble-based ML Model to Predict Cardiac Disease](#). In 2024 International Conference on Inventive Computation Technologies (ICICT) 2024 Apr 24 (pp. 214-219). IEEE
- C1 Rafi MH, **Mahjabin MR**, Rahman MS, Kabir MH, Ahmed S. [A Critical Analysis of Deep Learning Applications in Crop Pest Classification: Promising Pathways and Limitations](#). In 2023 26th International Conference on Computer and Information Technology (ICIT) 2023 Dec 13 (pp. 1-6). IEEE.

Professional Experience

University of South Florida , Tampa, FL	
Graduate Research and Teaching Assistant, Dept. of Computer Science and Engineering	Jan '24 - Current
Military Institute of Science and Technology (MIST) , Dhaka, Bangladesh	
Research Assistant, Dept. of Computer Science and Engineering	Aug '23 - Dec '23
Samsung R&D Institute , Dhaka, Bangladesh	
Software Engineer Intern	May '22 - Sept '22

Teaching

University of South Florida

Teaching Assistant

• CAI5133: Social Media Mining	Spring '25
• COP4530: Data Structures	Summer '24
• CIS4250: Ethical Issues/ Profess Conduct	Spring, Fall '24

Selected Projects

PPD Coach

Mobile Application

An android application for Screening Postpartum Depression in the context of Bangladesh	2023
• It screens postpartum depression using story-based questions and visual aids.	
• Developed with: <u>Flutter, Firebase</u>	

Software Defect Prediction

Notebook

Software fault prediction using machine learning [Github]	2022
• It predicts defects in software modules using different ML models	
• Developed with: <u>Python, Pandas, Numpy</u>	

HaateKhorī

Web Application

ML-powered web app for early childhood education [Github]	2021
• This project utilizes ML techniques to provide early education to children in an engaging and interactive way [Video Demo]	
• Developed with: <u>Python, OpenCV, YOLO, React, Nodes.js</u>	

Covid-CXRNet

Notebook

Detecting Covid19 with Chest X-ray using CNN [Github]	2021
• This project project implements a CNN model(ResNet18) to accurately detect COVID-19 with an accuracy of 99.98%	
• Developed with: <u>Python, Pytorch</u>	

Interests

Computational Social Science, Social Media Mining, Machine Learning, Data Science, Network Science, Affective Computing, Data Mining, Human-Computer Interaction, Natural Language Processing, Statistical Modeling

Skills

Programming Python, R, C/C++, Javascript, Java, HTML, CSS

Libraries PyTorch, TensorFlow, Keras, OpenCV, NLTK, Pandas, Scikit-Learn, Matplotlib, Numpy

Web & DB Flask, Django, React, MongoDB, Firebase, REST API

Tool Docker, Heroku, Git, LaTeX